QBITEL Bridge: Quantum-Safe Network Security in a Package

QBITEL Bridge offers AI-powered protocol discovery and post-quantum cryptography for legacy infrastructure. Enterprises must weigh adopting it to safeguard their systems.

Most enterprise networks have a dirty secret: nobody knows exactly what is running on them.

A hospital network built over 20 years has COBOL applications talking to modern REST services, HL7 medical devices communicating over undocumented serial-over-TCP wrappers, and PACS imaging systems using protocols nobody documented before the original vendor went bankrupt. A bank's mainframe infrastructure processes crores of rupees per second over NEFT/RTGS using IBM SNA protocol extensions that predate the internet.

And all of it is encrypted — if it is encrypted at all — using RSA or ECDH. These are algorithms that a sufficiently powerful quantum computer will break.

Two converging crises are forcing organisations to act: the unknown protocol problem (you cannot secure what you cannot see) and the harvest-now-decrypt-later threat (adversaries are archiving your encrypted traffic today, waiting for quantum hardware to mature).

QBITEL Bridge is an enterprise-grade open source platform, released under the Apache 2.0 licence, built to solve both — simultaneously, and without requiring any changes to existing applications or infrastructure.

What QBITEL Bridge does

At its core, QBITEL Bridge does four things:

- **Automatically discovers every protocol** on your network — including undocumented and legacy ones — using a hybrid AI ensemble of CNN, BiLSTM, and Transformer models.
- **Wraps all discovered protocols in NIST Level 5 post-quantum cryptography**, transparently, at the network layer.
- **Deploys autonomous AI agents** that monitor, classify, and respond to security events in under one second without human intervention.
- **Translates legacy protocols into modern REST/gRPC APIs**, auto-generating OpenAPI 3.0 specifications and SDKs in six languages.

The system sits between your existing infrastructure as a transparent bump-in-the-wire. It requires zero changes to applications, zero agent retraining, and zero downtime during deployment.

How the AI discovers unknown protocols

The most technically novel part of QBITEL Bridge is its protocol discovery subsystem. Traditional network security